Clase 4 (23/11/2025) EAFIT TDA

* Complejos simpliciales: e.g. $R_\alpha(X,d_X) \leftarrow$ complejo de Rips
* Característica de Euler: $\chi(K) = \sum_{n=0}^{\infty} (-1)^n (\#\ n\text{-simplejos de } K)$
* n-homología: $\preceq \leftarrow$ buen orden en vertices de K

$$\cdots \to C_{n+1}(K;\mathbb{F}) \xrightarrow{\partial_{n+1}} \underset{\underset{\text{Span}_{\mathbb{F}}(n\text{-simplejos de } K)}{\shortparallel}}{C_n(K;\mathbb{F})} \xrightarrow{\partial_n} C_{n-1}(K;\mathbb{F}) \to \cdots$$

frontera

$$\partial_n(\{x_0,\dots,x_n\}) = \sum_{i=0}^{n} (-1)^i \{x_0,\dots,x_{i-1},x_{i+1},\dots,x_n\}$$

$x_i \preceq x_j \Leftrightarrow i \leq j$

$$H_n(K;\mathbb{F}) = \frac{\ker(\partial_n)}{\text{img}(\partial_{n+1})}$$

$\ker(\partial_n) = Z_n(K;\mathbb{F}) \leftarrow$ ciclos

$\text{img}(\partial_{n+1}) = B_n(K;\mathbb{F}) \leftarrow$ fronteras

H_n: n-homología

Def: Numeros de Betti

$$\beta_n(K;\mathbb{F}) = \dim_{\mathbb{F}}(H_n(K;\mathbb{F})) \leftarrow \text{sobre } \mathbb{F}$$

$$\beta_n(K) = \dim_{\mathbb{R}}(H_n(K;\mathbb{R}))$$

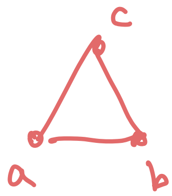
Importante: Si V es un espacio vectorial de dimensión finita y $W \leq V$, entonces $\dim(V/W) = \dim(V) - \dim(W)$

$$\Rightarrow \dim_{\mathbb{F}}\big(H_n(K;\mathbb{F})\big) = \dim_{\mathbb{F}}(\ker(\partial_n)) - \dim_{\mathbb{F}}(\operatorname{img}(\partial_{n+1}))$$

$$\beta_n(K;\mathbb{F}) = \dim_{\mathbb{F}}\big(H_n(K;\mathbb{F})\big)$$

$$= \underset{\text{nulidad}}{\operatorname{null}(\partial_n)} - \underset{\text{rango}}{\operatorname{rank}(\partial_{n+1})}$$

Ejemplos:

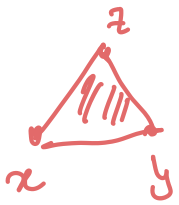


0-simplices

1-simplices

$$K_1 = \{\{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}\}$$

1 componente arco-conexa, 1 hueco $(ab + bc - ac)$



$$K_2 = \{x, y, z, xy, xz, yz, xyz\}$$

1 componente arco-conexa, no huecos

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Homología de K_1: Orden en vértices: $a \leq b \leq c$

$$\cdots \xrightarrow{\partial_3} C_2(K_1;\mathbb{F}) \xrightarrow{\partial_2} C_1(K_1;\mathbb{F}) \xrightarrow{\partial_1} C_0(K_1;\mathbb{F}) \xrightarrow{\partial_0} 0$$

$$\cdots \xrightarrow{\partial_3} \operatorname{Span}_{\mathbb{F}}(\text{triangulos}) \xrightarrow{\partial_2} \operatorname{Span}_{\mathbb{F}}(\text{aristas}) \xrightarrow{\partial_1} \operatorname{Span}_{\mathbb{F}}(\text{vértices}) \xrightarrow{\partial_0} 0$$

$$\cdots \xrightarrow{\partial_3} 0 \xrightarrow{\partial_2} \operatorname{Span}_{\mathbb{F}}(ab, ac, bc) \xrightarrow{\partial_1} \operatorname{Span}_{\mathbb{F}}(a, b, c) \xrightarrow{\partial_0} 0$$

$$ab \longmapsto b - a$$

$$ac \longmapsto c - a$$

$$bc \longmapsto c - b$$

$$* \ \mathrm{null}(\partial_0) = \dim_{\mathbb{F}}(\ker(\partial_0)) = \dim_{\mathbb{F}}(C_0(K_1;\mathbb{F})) = 3$$

$* \ 1 = \#$ componentes arco-conexas de $K_1 \leftarrow$ grafo

$$= \underbrace{\dim_{\mathbb{F}}(H_0(K_1;\mathbb{F}))}_{\beta_0(K_1;\mathbb{F})} = \mathrm{null}(\partial_0) - \mathrm{rank}(\partial_1)$$

$$= 3 - \mathrm{rank}(\partial_1)$$

$$\Rightarrow \boxed{\mathrm{rank}(\partial_1) = 2}$$

<u>Teorema</u> (rango-nulidad) Si $T: V \to V'$ es una transformación lineal, entonces

$$\boxed{\dim(V) = \mathrm{rank}(T) + \mathrm{null}(T)}$$

$$3 = \dim_{\mathbb{F}}(C_1(K_1;\mathbb{F})) = \mathrm{rank}(\partial_1) + \mathrm{null}(\partial_1)$$

$$= 2 + \mathrm{null}(\partial_1)$$

$$\Rightarrow \boxed{\mathrm{null}(\partial_1) = 1}$$

$\ker(\partial_1) = \mathrm{Span}_{\mathbb{F}}(ab + bc - ac)$ hueco (1-dim)

* $\dim_{\mathbb{F}}(H_1(K_1;\mathbb{F})) = \mathrm{null}(\partial_1) - \mathrm{rank}(\partial_2)$

$\quad\ \beta_1(K_1;\mathbb{F}) \quad = \quad 1 \ - \ 0 \quad = \quad 1$

Homología de K_2: $\quad x \leq y \leq z$

$$0 \xrightarrow{\partial_3} C_2(K_2;\mathbb{F}) \xrightarrow{\partial_2} C_1(K_2;\mathbb{F}) \xrightarrow{\partial_1} C_0(K_2;\mathbb{F}) \xrightarrow{\partial_0} 0$$

$$0 \xrightarrow{\partial_3} \mathrm{Span}_{\mathbb{F}}(xyz) \xrightarrow{\partial_2} \mathrm{Span}_{\mathbb{F}}(xy, xz, yz) \xrightarrow{\partial_1} \mathrm{Span}_{\mathbb{F}}(x, y, z) \xrightarrow{\partial_0} 0$$

$$xy \longmapsto y - x$$
$$yz \longmapsto z - y$$
$$xz \longmapsto z - x$$

$$xyz \longmapsto xy + yz - xy \longmapsto 0$$

* $1 = \dim(H_0(K_2;\mathbb{F})) = \beta_0(K_2;\mathbb{F}) = \#$ componentes arco-conexas

$= \mathrm{null}(\partial_0) - \mathrm{rank}(\partial_1) \Rightarrow \mathrm{rank}(\partial_1) = 2$

$\Rightarrow \mathrm{null}(\partial_2) = 1$

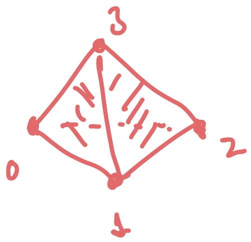
$\Rightarrow \ker(\partial_1) = \mathrm{Span}_{\mathbb{F}}(xy + yz - xz) = \mathrm{img}(\partial_2)$

$\Rightarrow \dim_{\mathbb{F}}(H_1(K_2;\mathbb{F})) = \mathrm{null}(\partial_1) - \mathrm{rank}(\partial_2)$

$\quad\ \beta_1(K_2;\mathbb{F}) = 0$ no huecos (1-dimensionales)

Ejemplo:

$$S = \mathcal{P}(\{0,1,2,3\}) - \{\emptyset, \{0,1,2,3\}\}$$



$\leftarrow$ hueco (sin 0123) en "dimensión 2"

\# huecos en dimensión 1 = 0

\# componentes arco-conexas = 1

$$\begin{tikzcd}
0 \arrow[r, "\partial_3"] & \mathrm{Span}_{\mathbb{F}}(012, 013, 023, 123) \arrow[r, "\partial_2"] & \mathrm{span}_{\mathbb{F}}(01, 02, 03, 12, 13, 23) \arrow[d, "\partial_1"] \\
 & 0 & \mathrm{Span}_{\mathbb{F}}(0, 1, 2, 3) \arrow[l, "\partial_0"']
\end{tikzcd}$$

matriz representando ∂_2 con respecto a la base de símplices

$$M_2 = \begin{array}{c} \\ 01 \\ 02 \\ 03 \\ 12 \\ 13 \\ 23 \end{array} \underbrace{\begin{array}{cccc} 012 & 013 & 023 & 123 \\ \left[\begin{matrix} 1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{matrix}\right] \end{array}}$$

$\leftarrow$ base espacio fila

$$\partial_2(012) = 01 + 12 - 02$$

$$\partial_2(013) = 01 + 13 - 03$$

$$\vdots$$

$$\mathrm{rank}(M_2) = \mathrm{rank}(M_2^T) = 3 \Rightarrow \boxed{\mathrm{null}(\partial_2) = 1}$$

$*\quad \dim_{\mathbb{F}}\left(H_2(S;\mathbb{F})\right) = \text{null}(\partial_2) - \text{rank}(\partial_3)$

$\qquad \| $

$\qquad \beta_2(S;\mathbb{F}) \qquad = 1 - 0 = 1 \leftarrow$ # huecos in dim 2

$*\quad \dim_{\mathbb{F}}\left(H_0(S;\mathbb{F})\right) = 1 \leftarrow$ # componentes arco-conexas

$\qquad \|$

$\qquad \beta_0(S,\mathbb{F}) \qquad = \text{null}(\partial_0) - \text{rank}(\partial_1)$

$$= 4 - \text{rank}(\partial_1)$$

$$\Rightarrow \text{rank}(\partial_1) = 3$$

$$\Rightarrow \boxed{\text{null}(\partial_1) = 3}$$

$*\quad \dim_{\mathbb{F}}\left(H_1(S;\mathbb{F})\right) = \text{null}(\partial_1) - \text{rank}(\partial_2)$

$\qquad \|$

$\qquad \beta_1(S;\mathbb{F}) \qquad = 3 - 3$

$= 0 =$ # huecos 1-dimensionales

$$\dim_{\mathbb{F}}\left(H_n(S;\mathbb{F})\right) = \begin{cases} 1, & n = 0 \\ 0, & n = 1 \\ 1, & n = 2 \\ 0, & n \geq 3 \end{cases}$$

$\uparrow$

triangula la 2-esfera  S^2

$\leftarrow$ no hay n-simplejos para $n \geq 3$

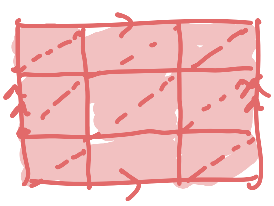
Idea: Si K es un complejo simplicial finito, entonces $H_n(K;\mathbb{F})$ es un espacio vectorial de dimensión finita

y $\beta_n(K;\mathbb{F}) = \dim_{\mathbb{F}}\left(H_n(K;\mathbb{F})\right) \sim$ # huecos n-dimensionales de K.

* $n=0 \leftarrow$ # de componentes arco-conexas
* $n=1 \leftarrow$ # de lazos cerrados no llenos (S^1)
* $n=2 \leftarrow$ # de cavidades (S^2)

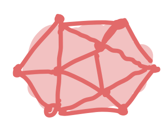
Otros ejemplos

1) El toro $T = S^1 \times S^1$

 $\cong$ 

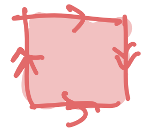
$$\beta_n(T;\mathbb{F}) = \begin{cases} 1, & n=0 \\ 2, & n=1 \\ 1, & n=2 \\ 0, & n \geq 3 \end{cases}$$

2) El plano Proyectivo real $\mathbb{R}P^2$



$$\beta_n(\mathbb{R}P^2;\mathbb{Z}_2) = \begin{cases} 1, & n=0 \\ 1, & n=1 \\ 1, & n=2 \\ 0, & n \geq 3 \end{cases} \quad ; \quad \beta_n(\mathbb{R}P^2;\mathbb{Z}_p) = \begin{cases} 1, & n=0 \\ 0, & n \geq 2 \end{cases}$$

$p \neq 2$ primo

3) La botella de Klein K  

$$\beta_n(K;\mathbb{Z}_2) = \begin{cases} 1, & n=0 \\ 2, & n=1 \\ 1, & n=2 \\ 0, & n\geq 3 \end{cases} \quad ; \quad \beta_n(K;\mathbb{Z}_p) = \begin{cases} 1, & n=0 \\ 1, & n=1 \\ 0, & n\geq 2 \end{cases}$$

$p \neq 2$ primo

4) La esfera m-dimensional $S^m \subseteq \mathbb{R}^{m+1}$

$$\beta_n(S^m;\mathbb{F}) = \begin{cases} 1, & n=0 \text{ o } n=m \\ 0, & \text{si } n\neq 0, m \end{cases}$$

Moraleja: $H_n(K;\mathbb{F})$ distingue los ejemplos que $\chi(\cdot)$ no.

Teorema: Si K es un complejo simplicial finito, entonces

$$\chi(K) = \sum_{n=0}^{\infty} (-1)^n \beta_n(K;\mathbb{F})$$

Prueba:

$$\sum_{n=0}^{\infty} (-1)^n \beta_n(K;\mathbb{F}) = \sum_{n=0}^{\infty} (-1)^n \left(\operatorname{null}(\partial_n) - \operatorname{rank}(\partial_{n+1}) \right)$$

$$= \sum_{n=0}^{\infty} (-1)^n \left(\dim_{\mathbb{F}}(C_n(K;\mathbb{F})) - \operatorname{rank}(\partial_n) - \operatorname{rank}(\partial_{n+1}) \right)$$

$$= \sum_{n=0}^{\infty} (-1)^n \cdot \dim_{\mathbb{F}} \left(C_n (K; \mathbb{F}) \right)$$

$$= \sum_{n=0}^{\infty} (-1)^n \left(\# \text{ de } n\text{-simplejos de } K \right)$$

$$= \chi(K) \;\blacksquare$$

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<u>En</u> <u>resumen</u>:

$$(X, d_X) \to \{ R_\alpha (X, d_X) \}_{\alpha > 0} \to \{ \beta_n (R_\alpha (X, d_X) ; \mathbb{F}) \}_{\alpha > 0}$$

$$\{ R_\alpha (X, d_X) \}_{\alpha > 0} \to \{ \chi (R_\alpha (X, d_X)) \}_{\alpha > 0}$$

datos — complejos simpliciales — invariantes

<u>Def</u>: Sea $\mathcal{K} = \{ K_\alpha \}_\alpha$ una familia de complejos simpliciales.

$$\alpha \longmapsto \beta_n (K_\alpha ; \mathbb{F}) \quad \text{curvas de Betti}$$

$$\alpha \longmapsto \chi (K_\alpha) \quad \text{curva carcteristica de Euler.}$$